ParkKumaon

Smart Parking Booking Platform for the Kumaon Region

Product Requirements Document (PRD)

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Version 2.0 — Updated with Acceptance Criteria, Edge Cases & North Star Metric
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1. Overview

Tourism in the Nainital, Haldwani, Bhimtal, and Kainchi Dham region has grown rapidly in recent years. These destinations attract thousands of tourists every day, especially during weekends, holidays, and peak tourist seasons. However, parking infrastructure and parking management systems have not scaled with the increasing tourist demand.

Visitors traveling by car often face the following issues:

- Difficulty finding available parking spaces
- Long waiting queues near tourist spots
- Traffic congestion caused by random parking
- Lack of real-time parking information
- No option to reserve parking in advance

These issues negatively impact the tourist experience, local traffic management, and environmental sustainability of the region.

To address this problem, we propose ParkKumaon, a digital parking booking platform that allows tourists and locals to discover, reserve, and pay for parking spots in advance near popular tourist destinations in the Kumaon region.

The platform will integrate:

- Real-time parking availability
- Advance parking reservation
- Digital payments
- Navigation to parking spots
- Local authority and parking vendor management

By digitizing parking infrastructure, ParkKumaon aims to:

- Reduce traffic congestion in hill towns
- Improve tourist convenience
- Help local authorities manage parking efficiently
- Enable local parking owners to monetize unused space

The platform will initially launch in high-traffic areas like Nainital, Bhimtal, Haldwani, and Kainchi Dham, and later expand across other tourist destinations in the Kumaon region.

This product aligns with the broader vision of smart tourism infrastructure, making travel smoother while supporting local businesses and governance.

2. Problem Statement

Tourist destinations in the Kumaon region such as Nainital, Haldwani, Bhimtal, and Kainchi Dham have witnessed a rapid increase in tourist inflow over the past decade. A significant proportion of tourists travel by private vehicles, which has dramatically increased the number of cars entering these hill towns. The region faces severe parking challenges due to limited infrastructure and lack of digital management systems.

Most parking areas operate through manual systems, and there is no centralized digital platform that allows tourists to identify or reserve parking spaces in advance. As a result, visitors arriving at popular locations such as Naini Lake often spend a considerable amount of time searching for parking.

Traffic Congestion

The absence of real-time parking information forces drivers to circulate repeatedly on narrow hill roads in search of available parking spaces, significantly increasing congestion in already capacity-constrained areas.

- Increased travel time for tourists and local residents
- Blocked emergency routes that can affect ambulances or disaster response
- Reduced mobility for local communities and businesses
- Higher fuel consumption due to vehicles idling or moving slowly in traffic

Environmental Impact

Continuous vehicle movement, long waiting times, and idling engines contribute to higher emissions of CO_2 , NOx, and particulate matter, which degrade air quality — particularly concerning near ecologically sensitive areas such as Naini Lake.

- Noise pollution, disturbing the natural environment and local residents
- Increased pressure on fragile mountain infrastructure
- Higher fuel consumption and carbon footprint of tourism

Inefficient Utilization of Parking Resources

Many private parking spaces owned by hotels, small businesses, or local landowners remain underutilized because they are not listed on any digital platform.

- Inefficient use of available parking infrastructure
- Lost revenue opportunities for local parking owners
- Continued congestion around major tourist spots

Lack of Digital Management

Most parking areas rely on manual ticketing and cash payments, leading to inefficiencies, lack of transparency, and difficulty in monitoring parking usage patterns.

Problem Summary

Issue	Description
No Parking Visibility	Tourists don’t know where parking spaces are available or what they cost.
Traffic Congestion	Vehicles circle around locations like Naini Lake searching for parking.
Time Waste	Visitors spend 20–60 minutes searching for parking.
Unorganized Parking	Many private parking spaces are not listed anywhere.
Manual Payment System	Cash-based parking leads to inefficiency and leakages.

Issue	Description
Tourist Experience Issues	Visitors often start their trip with frustration.

Core Problem

The core problem is the absence of a centralized digital system that provides real-time parking availability, advance reservation capabilities, and efficient parking management in major tourist destinations of the Kumaon region.

Without such a system, tourists will continue to face parking uncertainty, traffic congestion, and time loss, while the region will experience increased environmental pressure and inefficient utilization of existing parking resources. A smart parking booking platform can address these issues by enabling tourists to discover, reserve, and navigate to parking spaces before reaching their destination.

2.A Competitor Analysis

Competitor	Strengths	Gaps
Park+	Provides car services and limited parking solutions	Does not focus on tourist hill regions
Google Maps	Helps in navigation	No parking booking functionality
Local Offline Parking	Existing manual system	Completely manual; no visibility or booking; inefficient and unorganized

Gap in Market:

- No platform provides advance parking booking in tourist destinations like Nainital and Bhimtal
- No system for local parking owners to list spaces

This creates a clear opportunity for the ParkKumaon platform.

3. Founder Vision & Motivation

As a frequent traveler to hill stations, I have personally experienced the frustration of finding parking in crowded tourist destinations like Nainital. This inspired the idea of building a smart parking platform that improves travel experience while supporting sustainable tourism in Uttarakhand.

4. Early Validation

Initial user discussions revealed that tourists spend 30–60 minutes searching for parking during peak seasons. Conversations with potential parking owners indicated interest in listing their spaces on a digital platform.

5. Goals & Objectives

The primary goal of the Smart Parking Booking Platform is to create an efficient, organized, and sustainable parking management system for major tourist destinations in the Kumaon region.

The platform aims to improve the overall travel experience for tourists, optimize the use of existing parking infrastructure, support local economic opportunities, and reduce the environmental impact caused by unmanaged vehicle traffic.

Primary Goals

1. Reduce Parking Search Time

By providing real-time parking availability and advance booking options, tourists will be able to identify and reserve parking before reaching their destination. This can potentially reduce parking search time by 50–70%.

2. Reduce Traffic Congestion

The platform aims to reduce congestion by directing vehicles to available parking spaces through digital navigation and booking systems, distributing vehicles across multiple parking locations.

3. Promote Environmental Sustainability

By enabling tourists to reserve parking in advance and reach it directly, the platform aims to reduce CO_2 emissions, reduce fuel consumption, improve air quality, and protect ecosystems around areas like Naini Lake.

4. Improve Tourist Experience

The platform aims to create a seamless, stress-free experience via parking discovery, advance reservations, digital payments, and navigation assistance.

5. Optimize Utilization of Existing Parking Infrastructure

Bringing underutilized hotel/commercial/private parking spaces onto a centralized marketplace, reducing pressure on popular lots and increasing availability.

6. Create Economic Opportunities for Local Stakeholders

Allowing local parking owners, businesses, and landowners to list spaces and generate additional revenue, turning unused land into a monetizable asset.

7. Enable Data-Driven Parking Management

Providing usage data insights to local authorities to monitor peak demand, identify congestion hotspots, and plan infrastructure and traffic policy.

Key Success Targets

- Reduce parking search time by 60–70%
- Achieve 30% booking conversion rate
- Maintain cancellation rate below 20%
- Achieve 80% parking slot utilization in peak areas

Long-Term Vision

The long-term vision of the platform is to transform parking management in tourist destinations across Uttarakhand by creating a smart, sustainable, and digitally integrated parking ecosystem, supporting better mobility, environmental conservation, and improved urban planning in hill regions.

Business Goals

- Create a commission-based marketplace for parking
- Partner with local parking owners
- Generate revenue through booking commission, listing fees, and premium parking

6. Impact on Uttarakhand Economy & Environment

- Reduce parking search time by 60–70%
- Reduce traffic congestion in peak tourist areas by 25–30%
- Increase parking space utilization up to 80%
- Generate additional income opportunities for local parking owners
- Reduce CO_2 emissions caused by idle traffic in eco-sensitive regions

This platform directly supports Uttarakhand's vision of smart and sustainable tourism infrastructure.

7. Target Users

User Type	Description	Key Needs
Tourists	People traveling to hill stations by car (e.g. family visiting Nainital)	Reserve parking, avoid traffic, secure parking
Local Residents	People visiting markets or offices in cities like Haldwani	Quick short-term parking
Parking Owners	Hotels, landowners, malls, or local authorities	Monetize empty land
Local Government / Municipality	City and tourism administration	Manage city traffic, monitor parking usage

8. User Stories

8.A Tourist Planning a Trip (Pre-Travel Experience)

User: Tourist traveling with family

As a tourist planning a trip to Nainital, I want to check available parking spaces online before leaving my city, so that I feel confident I will find parking easily when I reach the destination.

User Experience

User opens the platform and sees a search bar: "Where do you want to park?" They enter a location and see a map of available spaces near major tourist spots. They select date and time, view a list of parking options with pricing, choose one, confirm and pay, and see the booking on their dashboard.

User Feels

- Relieved

- Confident about the trip

Acceptance Criteria

- Given a valid location is entered, the system displays parking results within 3 seconds.
- Given no parking is available near the searched location, the system shows the nearest alternative within 2 km.
- Given the user selects date/time, only real-time available slots for that window are shown.

8.B Tourist Booking Parking Before Arrival

User: Visitor driving to a temple or tourist destination

As a visitor going to Kainchi Dham, I want to reserve a parking slot in advance, so that I do not get stuck in traffic or struggle to find parking during peak hours.

User Experience

The platform shows parking options with price, distance, and available slots. The user selects a spot 300m from the temple and books it for 10 AM–2 PM. After payment, the user receives a digital confirmation and QR code.

User Feels

- Stress-free
- Prepared for the journey
- Confident about reaching the destination without parking issues

Acceptance Criteria

- Given a user selects a slot, date, and time, the system locks the slot for 10 minutes pending payment.
- Given payment is successful, a QR code and confirmation (SMS/email) are generated within 30 seconds.
- Given two users attempt to book the same last slot simultaneously, only one booking succeeds; the other sees “Slot just got booked” and is shown the next nearest option.

8.C Tourist Arriving at Destination

User: Tourist arriving by car

As a tourist arriving in Bhimtal, I want clear navigation to my booked parking location, so that I can reach it easily without creating traffic or confusion.

User Experience

On reaching the city, the platform automatically opens map navigation guiding the driver directly to the reserved parking lot.

User Feels

- Convenience
- Time saved
- Less travel stress

Acceptance Criteria

- Given a booking is confirmed, in-app navigation auto-launches when the user is within 5 km of the parking location.

- Given GPS signal is weak (hill terrain), the app falls back to last-known cached map directions.

8.D Tourist Concerned About Environment

User: Responsible traveler

As a tourist who cares about environmental sustainability, I want to avoid unnecessary driving while searching for parking, so that I can reduce fuel consumption and environmental impact in eco-sensitive areas like Kumaon.

User Experience

The platform helps the user drive directly to the reserved parking location, avoiding unnecessary traffic and reducing fuel consumption.

User Feels

- Contributing to sustainable tourism
- Protecting the natural environment

8.E Local Resident Visiting City Market

User: Local resident in Haldwani

As a local resident visiting the market area, I want to quickly find available parking near my destination, so that I do not waste time searching for parking in busy areas.

User Experience

The user opens the platform and sees nearby parking options with real-time availability, chooses a spot, and navigates to it within seconds.

User Feels

- Convenience
- Efficiency
- Better city mobility

8.F Parking Owner Listing Their Parking Space

User: Local landowner or hotel owner

As a parking owner in Nainital, I want to list my available parking space on the platform, so that I can earn additional income from tourists visiting the area.

User Experience

The owner logs in and adds location, number of slots, and price per hour. Once approved, the listing becomes visible to tourists.

User Feels

- Empowered
- Financially benefited
- Connected to the tourism ecosystem

Acceptance Criteria

- Given an owner submits a listing, it enters “Pending Approval” status and is not publicly visible until admin approval.

- Given a listing is approved, it appears in search results within 5 minutes.

8.G Local Authority Monitoring Parking

User: City administrator or municipal authority

As a city administrator managing tourism traffic in Nainital, I want to monitor parking usage and demand, so that I can manage traffic and plan better infrastructure.

User Experience

Through the admin dashboard, authorities can see parking occupancy, peak traffic hours, and most-used parking locations.

User Feels

- Informed
- Better equipped for infrastructure planning

8.H Booking Cancellation Experience

User: User with a confirmed booking

As a user, I want to cancel my parking booking if my travel plan changes, so that I don't lose my entire booking amount.

User Experience

The user opens the dashboard, selects their booking, and clicks "Cancel Booking." The system clearly shows the cancellation charge and refund amount before confirming.

User Feels

- Transparency
- Control
- Trust in the platform

Acceptance Criteria

- Given cancellation occurs ≥ 4 hours before booking time, charge is 20% of booking amount or ₹20, whichever is higher.
- Given cancellation occurs < 4 hours before booking time, charge is 70% of booking amount.
- Given a refund is due, it is processed to the original payment method within 3–5 business days.

Overall User Feeling

- Convenience – easy parking discovery
- Confidence – guaranteed parking availability
- Time efficiency – reduced travel delays
- Environmental responsibility – reduced unnecessary driving
- Better travel experience in destinations like Nainital and Bhimtal

9. Key Features

9.A User Registration & Login

Description

Users must be able to create an account to book parking spaces.

Functional Requirements

- Sign up using email or phone number
- Login with OTP
- Reset password
- View profile information

Why This Feature is Important

- Saves booking history
- Enables quick bookings
- Allows personalized recommendations

Acceptance Criteria

- Given a valid phone number, OTP is delivered within 30 seconds.
- Given 3 incorrect OTP attempts, the field locks for 5 minutes.
- Given a user requests password reset, a reset link/OTP expires after 10 minutes.

9.B Search Parking

Description

Users should be able to search parking based on location (e.g. near Naini Lake, Kainchi Dham, Bhimtal).

Functional Requirements

- Filters: Location, Distance, Price, Availability, Parking type

Acceptance Criteria

- Given a search query with no exact match, the system suggests the 3 nearest results.
- Given filters are applied, results update without a full page reload.

9.C Real-Time Parking Availability

Description

Users must see live parking slot availability (e.g. Parking Lot A — Total Slots: 50, Available: 12).

Why This Feature is Important

- Prevents traffic congestion
- Prevents time wasted searching for parking

Acceptance Criteria

- Given a slot is booked by another user, availability count updates within 5 seconds across all sessions.
- Given the data source (owner update) is delayed beyond 2 minutes, the listing shows a “last updated” timestamp.

9.D Parking Slot Booking

Description

Users can reserve parking in advance. Booking flow: open platform → search location → select parking → choose time slot → confirm booking.

Acceptance Criteria

- Given a user confirms a slot, the system prevents double-booking of the same slot for an overlapping time window.
- Given booking confirmation is generated, it includes Booking ID, location, time window, and price.

9.E Online Payment

Description

Users must pay digitally via UPI, Debit Card, Credit Card, or Wallet. Recommended gateways: Razorpay, Stripe.

Acceptance Criteria

- Given payment fails after deduction, the amount is auto-reversed within 24 hours and the booking is not confirmed.
- Given payment succeeds, booking status updates to “Confirmed” before the user is redirected to the confirmation page.

9.F Navigation to Parking

Description

After booking, users get Google Maps navigation directly to the reserved spot.

Why This Feature is Important

- No confusion
- Direct route
- Reduced traffic

Acceptance Criteria

- Given the user taps “Navigate,” the app opens turn-by-turn directions to the exact parking entrance, not just the area.

9.G QR Code Entry

Description

At the parking gate, the user shows a QR code which the operator scans.

Why This Feature is Important

- Fast entry
- No paper tickets
- Digital tracking

Acceptance Criteria

- Given a QR code is scanned, entry is logged with a timestamp and the slot is marked “Occupied.”
- Given a QR code is scanned twice, the second scan is rejected with a duplicate-entry alert.

9.H Parking Owner Dashboard

Description

Owners can add parking spaces, update slot availability, set price, view bookings, and track revenue.

Acceptance Criteria

- Given an owner updates availability, the change reflects in search results within 5 minutes.

9.I Admin Dashboard

Description

Admin manages the platform: approve listings, monitor usage, view traffic analytics, override bookings in emergencies, block suspicious users, control dynamic pricing.

9.J Booking Cancellation & Refund Policy

Description

Users can cancel a booking before the scheduled arrival time, subject to a cancellation policy that ensures fair usage and prevents last-minute slot blocking.

Cancellation Rules

- Cancellation ≥ 4 hours before booking time → Charge: 20% of booking amount or ₹20 (whichever is higher)
- Cancellation < 4 hours before booking time → Charge: 70% of booking amount

Refund Policy

- Remaining amount refunded to user's original payment method
- Refund processed within 3–5 business days

No-Show Policy

- If user does not arrive within 1 hour of booking time, the booking is marked No-Show and no refund is provided.

Why This Feature is Important

- Prevents misuse of the booking system
- Ensures parking slot availability
- Compensates parking owners for last-minute cancellations
- Encourages responsible booking behavior

Acceptance Criteria

- Given a cancellation request, the exact charge and refund amount are displayed before the user confirms cancellation.
- Given a No-Show, the slot is auto-released for other users 1 hour after the booked start time.

9.K Trust & Safety

Description

Ensures user trust and platform reliability.

Functional Requirements

- Verified parking owners before listing
- User ratings & reviews for parking locations
- Secure payment processing
- Fraud detection for fake bookings
- Customer support for dispute resolution

Why This Feature is Important

- Builds user trust
- Improves platform credibility
- Reduces misuse and fraud

9.L Notifications & Alerts

Description

Keeps users informed throughout the booking lifecycle.

Functional Requirements

- Booking confirmation (SMS/Email)
- Reminder before parking time
- Cancellation confirmation
- Refund updates

10. Functional Requirements — Platform Pages

10.A Home Page

Purpose: Introduce the platform.

- Search parking bar
- Map showing parking spots
- Top destinations: Nainital, Bhimtal, Kainchi Dham

10.B Search Results Page

- List of parking spaces
- Map view
- Price
- Distance
- Available slots

10.C Parking Details Page

- Parking location
- Photos
- Price
- Total slots

- Available slots
- Distance from tourist spot (e.g. Parking near Naini Lake)

10.D Booking Page

User selects Date, Time, and Vehicle type (e.g. car parking for 4 hours).

- Show cancellation & refund policy clearly before payment confirmation
- User must accept policy before booking

10.E Payment Page

Payment confirmation page shows Booking ID, QR Code, and Parking location, along with the refund breakdown applicable in case of cancellation.

10.F User Dashboard

- Booking history
- Upcoming bookings
- Saved locations

Booking Cancellation Flow

1. User goes to Dashboard
2. Selects Upcoming Booking
3. Clicks Cancel Booking
4. System calculates cancellation charge and shows: Booking Amount ₹500, Cancellation Charge ₹100, Refund Amount ₹400
5. User confirms cancellation
6. Booking status → Cancelled, refund initiated

10.G Parking Owner Dashboard

- Parking slots
- Pricing
- Earnings

11. Non-Functional Requirements

- Fast loading (<2 seconds)
- Secure payment gateway
- Mobile responsive
- GPS integration
- Cloud hosting

12. Tech Stack (Recommended)

Layer	Recommendation
Frontend	React.js, Next.js
Backend	Node.js, Express
Database	MongoDB / PostgreSQL
Maps Integration	Google Maps API
Payment Gateway	Razorpay, Stripe
Hosting	AWS, Vercel

13. MVP (Minimum Viable Product)

For launch you only need:

- User login
- Parking search
- Parking listing
- Booking system
- Online payment
- Map navigation

Cities for MVP:

- Nainital
- Haldwani

MVP Feature Prioritization

Must Have Features:

- User login
- Parking search
- Parking booking
- Online payment
- Basic navigation
- Basic cancellation support

Good to Have Features:

- QR code entry
- Admin analytics dashboard
- Parking owner advanced dashboard

Future Features:

- Smart parking sensors
- Dynamic pricing

13.A Current Product Stage

The product is currently at the prototype stage with a demo website validating the core user journey (search, booking, payment). The next phase involves building a functional MVP and onboarding initial parking partners.

14. Monetization Model

Commission Model

Take 10–20% per booking.

Listing Fee

Parking owners pay a monthly listing fee.

Cancellation Revenue

The platform can generate revenue from cancellation charges — early cancellation incurs a small fee, late cancellation a higher fee. This increases revenue, reduces fake bookings, and improves slot utilization.

Premium Parking

VIP parking near high-demand locations such as Naini Lake.

14.A Use of Funds

Allocation	Amount
Product development	₹2 lakh
Marketing	₹1 lakh
Operations & partnerships	₹1 lakh
Team support	₹1 lakh

15. Risks & Challenges

Risk	Mitigation
Low Parking Owner Adoption	Incentives and zero listing fee initially.
Real-Time Data Accuracy	IoT sensors or manual updates.
Network Issues in Hills	Offline booking confirmation via SMS.
High Cancellations by Users	Apply structured cancellation charges; show policy clearly before booking.
Users May Not Trust Advance Booking Initially	Offer first-booking discount; enable easy cancellation; show real-time availability.

15.A Edge Cases

Consolidated list of edge cases the engineering and QA teams must explicitly design and test for, mapped to impact and proposed handling.

#	Edge Case	Impact	Proposed Solution
1	User does not arrive after booking	Slot blocked unnecessarily; owner loses revenue	No refund after booking time; slot auto-released after 1 hour (No-Show policy)
2	Parking shown available but is physically full on arrival	Loss of trust; user stranded	Backup parking allocation at nearby lot, or full refund + apology credit
3	Network issues in hill areas during booking/payment	Failed transactions; user unable to confirm	Offline booking confirmation via SMS; retry queue for payment confirmation
4	User cancels bookings repeatedly (abuse pattern)	Revenue loss; blocks slots for genuine users	Limit number of free/low-charge cancellations per user per month
5	Payment deducted but booking not confirmed (gateway timeout)	User loses money with no booking	Auto-reconciliation job; auto-refund within 24 hours if booking not created
6	Parking owner reduces or removes listed slots after bookings are already confirmed	Overselling; confirmed users have no slot	Lock booked slots from owner-side edits; require admin approval for slot reduction
7	Two users book the last available slot at the same instant	Double booking, only one can be honored	Slot-locking mechanism (10 min hold) + atomic booking transaction at confirmation
8	GPS/navigation shows incorrect parking entrance due to poor hill-area map data	User confusion, wasted time, traffic	Manual entrance-point tagging by owner; fallback to landmark-based directions
9	User books multiple vehicles under one account for the same slot/time	Potential fraud or scalping	One active booking per vehicle number per time window; flag for review
10	Refund initiated but delayed or sent to wrong account	User dissatisfaction; support load	Refund status visible in dashboard; escalation path after 5 business days

16. Timeline / Milestones

Month	Milestone
Month 1	Research + Design

Month	Milestone
Month 2	MVP development
Month 3	Beta launch in Nainital
Month 4	Expansion to Bhimtal and Haldwani

16.A Go-To-Market Strategy

7. Partner with parking owners in Nainital and Haldwani
8. Collaborate with hotels and travel agencies
9. Promote via social media and Google search ads
10. Provide free listing for parking owners initially
11. Offer discounts for first-time users

Goal:

Acquire first 1,000 users in the Nainital region.

17. Future Features

- Smart parking sensors
- Dynamic pricing
- EV charging parking
- Parking subscription
- Tourist itinerary integration
- Small hotel booking integration
- Authentic real-time road mapping using the NAVIC GPS system

18. Success Metrics

North Star Metric

% of Searches Converted into a Friction-Free Completed Booking

Defined as: the percentage of parking searches that result in a completed, paid booking where the user finds and reserves a slot in under a target search time threshold (e.g. under 3 minutes). This single metric captures both core value delivered to the user (fast, successful parking discovery) and business growth (booking volume, the platform's revenue driver) — directly tying back to the core problem statement in Section 2.

Rationale: a pure booking-count metric could be gamed by low-quality, slow bookings. A pure speed metric ignores whether the user actually completed a transaction. Combining both ensures the team optimizes for genuine problem-solving, not vanity growth.

Supporting / Input Metrics

Metric	Definition	Target
Average Parking Search Time	Time from search initiated to slot selected	Reduce by 60–70%
Booking Conversion Rate	% of users who complete booking after search	30%+
Daily Active Users (DAU)	Number of users using the platform daily	Growing MoM
Cancellation Rate	% of confirmed bookings that get cancelled	< 20%
Revenue per Booking	Average platform earnings per booking	Track & optimize
Parking Utilization Rate	% of listed slots booked vs. total available	80% in peak areas

These supporting metrics will be tracked weekly on the admin dashboard and used to diagnose movement (or lack of movement) in the North Star Metric.